Graph Machine Learning

Multi-Relational Graph Embedding Approaches

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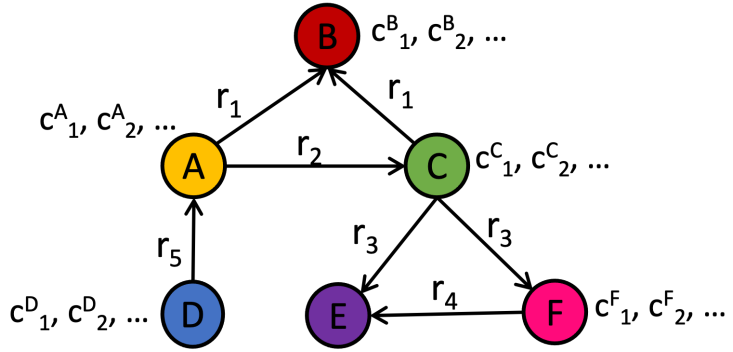
Multi-relational and Knowledge Graphs

- Multi-relational graph refers to a directed graph $\mathcal{G}(E, R)$ whose nodes are entities $\in E$ and edges are relationships between entities $\in R$.
- $\mathcal{G}$ could be represented as a set of triples of head h, tail t and their relationship r

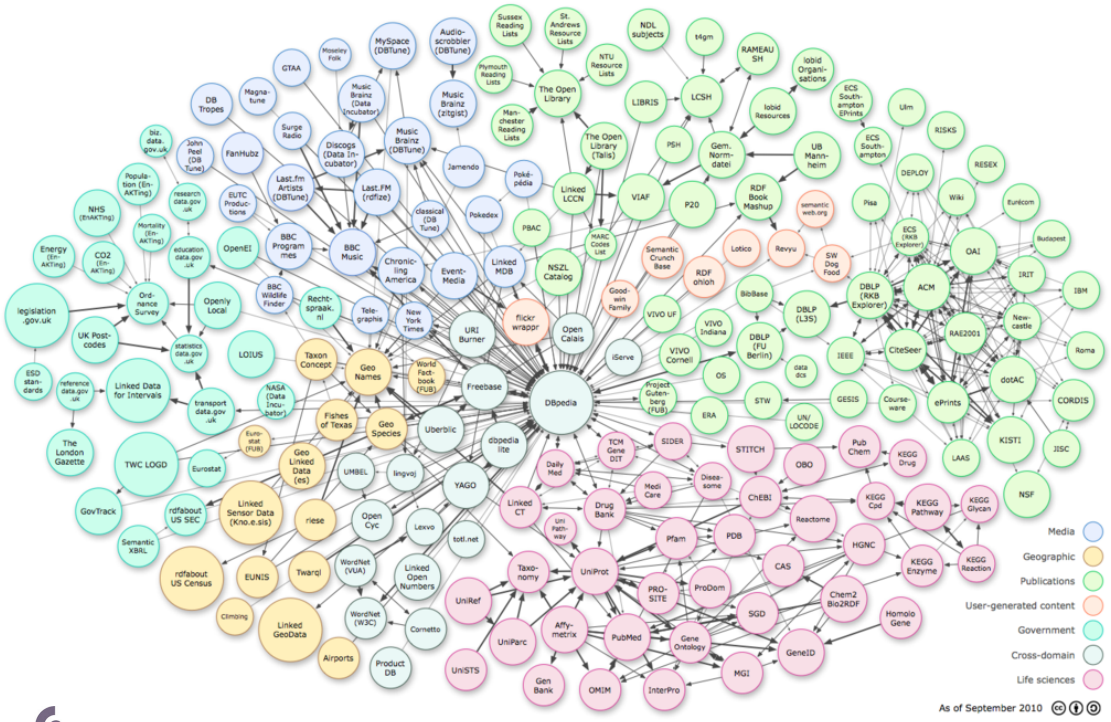
$$\{(h, r, t) | h, t \in E \text{ and } r \in R\}$$

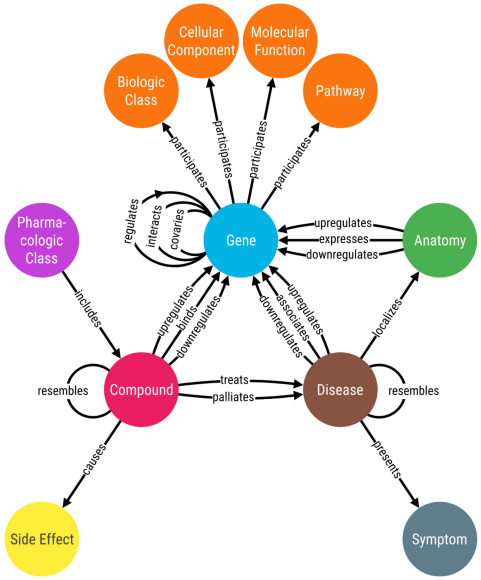
Nodes: entities of different classes c_1, c_2, ...
Edges: relationships between entities of different types: r_1, r_2, r_3, r_4 and r_5



Multi-relational and Knowledge Graphs

- Examples of multi-relational data:
  → Social network (Facebook, LinkedIn) , Recommender systems (Amazon Product Graph), . . .
  → Knowledge Graphs (KGs): DBPedia, Google KG, IBM Watson, Microsoft Satori, Freebase, Yago, Nell or GeneOntology, Some of them are publicly available (linked open data)
- KG: multi-relational graph where the semantics underlying the entities and their relationships (Abox, Assertion Box) are formally described in the Tbox (Terminology Box)
  → reasoning (inference) capabilities based on logical facts (axioms and rules)
  → W3C languages for representation and querying such graphs





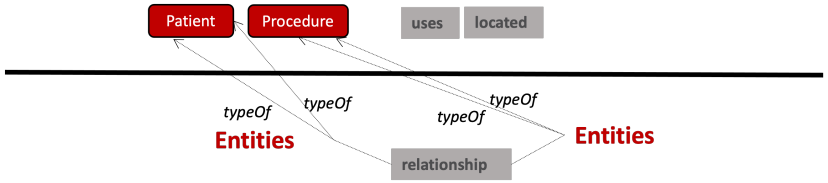
Multi-relational and Knowledge Graphs

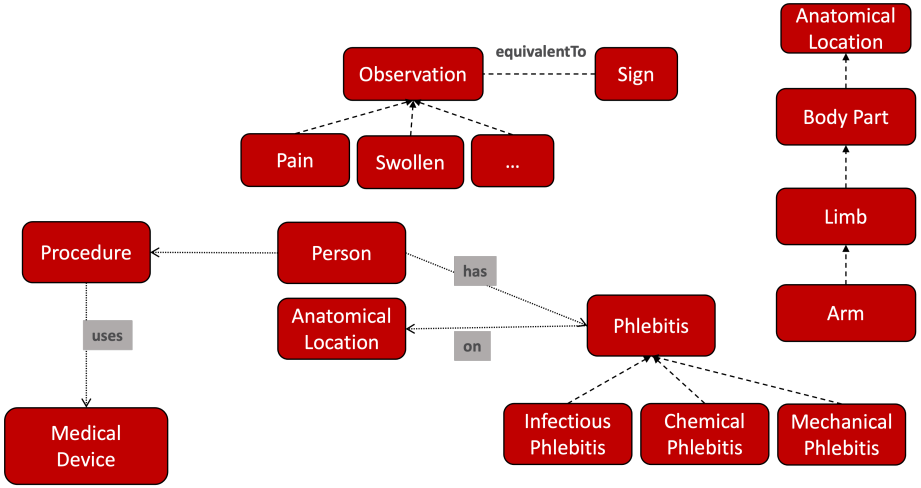
☞ KG: multi-relational graph where the semantics underlying the entities and their relationships (Abox, Assertion Box) are formally described in the Tbox (Terminology Box)

→ Some examples: classes of entities, subclassOf between classes, domain/range of properties, subproperty between properties, symmetry, transitivity, composition, 1-to-Many, ...

A knowledge a graph-structured data to represent entities, their relationships while also encoding the semantics underlying these entities.

Classes, Properties Triples + Rules





Multi-relational and Knowledge Graphs

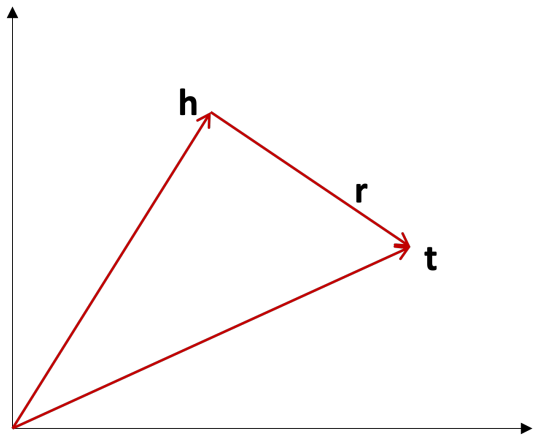
- ☞ Applications: Question answering/conversational agents, recommendation systems, search engines, . . .
- ☞ Common characteristics: massive, noisy and incomplete
- ☞ Problem: how to embed entities and relationships of multi-relational data in low-dimensional vector spaces. How to define similar entities using triples?
- ☞ Translational approaches TransE and extensions ...

Multi-relational Graph Embedding: TransE

- ☞ Translational methods (also shallow embedding) such as TansE, TransH, TransR, . . . and many others DistMult, ComplEx, and RotatE, TransOWL
- ☞ TransE (Bordes et al.) canonical model, easy to train, with low number of parameters and scale up to very large KGs.
- ☞ TransE represents both entities and relations as vectors in the same space, $\mathbb{R}^d$, d is a hyperparameter. Every entity head h and tail t and every relation (r) is represented as a low-dimensional vector (e.g., d 50 to 200 dimensions).

Multi-relational Graph Embedding: TransE

- ✓ Given $(h \in E, r \in R, t \in E)$, the relation embedding $\mathbf{r}$ is interpreted as a translation vector from head embedding $\mathbf{h}$ to tail embedding $\mathbf{t}$ *st.* $\mathbf{t} \approx \mathbf{h} + \mathbf{r}$
- ✓ For $(\mathit{AlfredHitchcock}, \mathit{directorOf}, \mathit{Psycho}), (\mathit{JamesCameron}, \mathit{directorOf}, \mathit{Avatar})$ the intuition is to capture regularities on $(\mathit{Psycho}, \mathit{AlfredHitchcock})$ and $(\mathit{Avatar}, \mathit{JamesCameron})$ to get $\mathit{AlfredHitchcock} + \mathit{directorOf} \approx \mathit{Psycho}$ and $\mathit{JamesCameron} + \mathit{directorOf} \approx \mathit{Avatar}$.

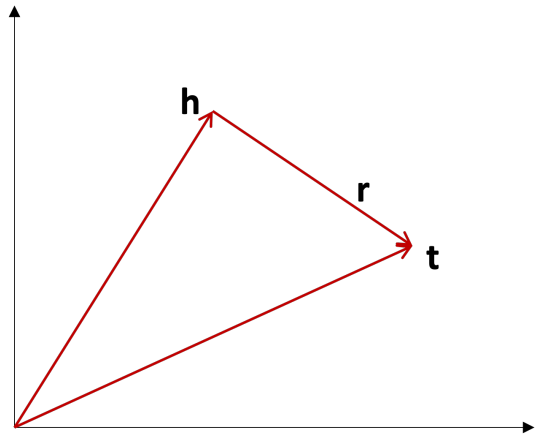


Multi-relational Graph Embedding: TransE

- ☞ TransE represents both entities and relations as vectors in the same space, $\mathbb{R}^d$, d is a hyperparameter.
- ☞ To measure how well a triple fits, the similarity score between **h** translated by **r** and **t**, using the norm L_2 is:

$$f_r(h,t) = ||\mathbf{h} + \mathbf{r} - \mathbf{t}||_2$$

  - ✓ A lower score means the triple is more likely to be true: (h, r, t) holds.
  - ✓ Squaring penalises large errors more heavily than small ones.



Multi-relational Graph Embedding: TransE

- ☞ TransE is trained using margin-based ranking loss. It doesn't just want to make "true" facts have low scores; it wants to make "false" facts have high scores.
- ☞ Given a training set $\mathcal{S}^+$ of triples $(h \in E, r \in R, t \in E)$, and the set $\mathcal{S}^-$ of corrupted triples constructed according to:

$$\mathcal{S}^-_{(h,r,t)} = \{(h\prime, r, t) | h\prime \in E\} \cup \{(h, r, t\prime) | t\prime \in E\}$$

  with either the head or tail replaced by a random entity (but not both at the same time).
- ☞ To learn such embeddings, a loss function, a margin-based ranking over the training set:

$$\mathcal{L} = \sum_{(h,r,t)\in\mathcal{S}^+} \sum_{(h\prime,r,t\prime)\in\mathcal{S}^-_{(h,r,t)}} \left[\gamma + f_r(h,t) - f_r(h\prime,t\prime)\right]$$

  The goal is to make the score of positive triples significantly better than negative ones.
  - ✓ $\gamma > 0$ is a margin hyperparameter.
  - ✓ Note that for a given entity, its embedding vector is the same when the entity appears as the head or as the tail of a triplet.
  - ✓ The optimization is carried out by SGD.

Multi-relational Graph Embedding: TransE

input: $\mathcal{S}^+ = \{(h, r, t)\}$, E, R, dimension d, margin γ
output: Embedding vectors $\mathbf{e}, \mathbf{r}$
initialization: Embedding vectors $\mathbf{e}, \mathbf{r}$ are randomly uniformly initialized, and normalised
$r \leftarrow uniform(\frac{-6}{\sqrt{d}}, \frac{6}{\sqrt{d}}), \mathbf{r} \leftarrow \frac{\mathbf{r}}{||\mathbf{r}||} r \in R, \mathbf{r} \in \mathbb{R}^{|R|}$
$e \leftarrow uniform(\frac{-6}{\sqrt{d}}, \frac{6}{\sqrt{d}}), e \in E$
while stop condition **do**
  $\mathbf{e} \leftarrow \frac{\mathbf{e}}{||\mathbf{e}||}, \mathbf{e} \in \mathbb{R}^{|E|}$ /**entity embedding norm is 1, to prevent to trivially minimize $\mathcal{L}$ by artificially increasing entity embeddings norms.**/
  $\mathcal{S}_{batch} \leftarrow sample(\mathcal{S}^+, b)$ /**From the training set randomly sample a batch of size b**/
  for each $(h, r, t) \in \mathcal{S}_{batch}$ **do**
    $(h', r, t') \leftarrow sample(\mathcal{S}^-)$ /**sample a corrupted fact of triple**/
    $T_{batch} \leftarrow T_{batch} \cup \{((h, r, t), (h', r, t'))\}$
  end for
  Update embeddings by minimizing the loss function using GDS
end while

Multi-relational Graph Embedding: TransE

✓ What is the behavior of TransE given the properties of the relations? Is TransE expressive enough to model these relations?

- ☞ Symmetry: $(h, r, t) \in \mathcal{G} \Rightarrow (t, r, h) \in \mathcal{G}$. Examples of r: roommate, brother, ...
- ☞ Inverse: $(h, r_1, t) \in \mathcal{G} \Rightarrow (t, r_2, h) \in \mathcal{G}$. Examples of r_1 and r_2: advisor and advisee, parent and child, husband and wife ...
- ☞ Composition: $(h, r_1, th), (th, r_2, t) \in \mathcal{G}$ and $\Rightarrow (h, r_3, t) \in \mathcal{G}$.
- ☞ Transitive: $(h, r, th), (th, r, t) \in \mathcal{G}$ and $\Rightarrow (h, r, t) \in \mathcal{G}$. Examples of r: sibling, ..
- ☞ 1-to-N: $(h, r, t_1) \in \mathcal{G} \Rightarrow \exists (h, r, t_2) \in \mathcal{G}$. Examples of r: mother, father ..
- ☞ Antisymmetry: $(h, r, t) \in \mathcal{G} \Rightarrow \nexists (t, r, h) \in \mathcal{G}$. Ex. : parent, ...

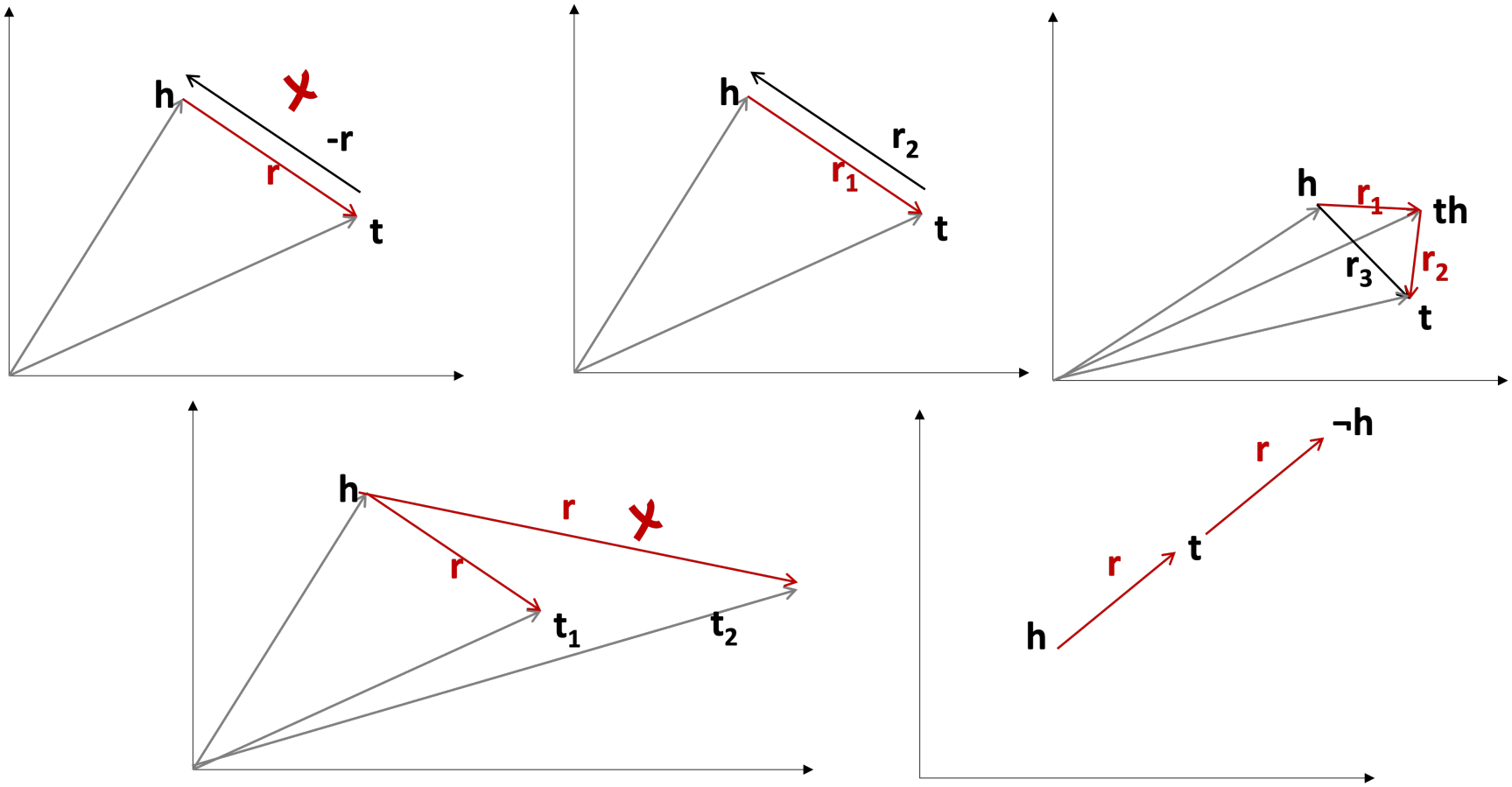
Multi-relational Graph Embedding: TransE

- ☞ **Symmetry**: $(h, r, t) \in \mathcal{G} \Rightarrow (t, r, h) \in \mathcal{G}$. Examples of r:
- ☞ $\mathbf{h} + \mathbf{r} = \mathbf{t}$ and $\mathbf{t} + \mathbf{r} = \mathbf{h}$, this forces $\mathbf{r}$ to be 0 vector.
- → In "mariedWith" relation, "husband" and "wife" entities are represented by the same point in the space. The model in not able to distinguish between them.
- ☞ **Inverse**: $(h, r_1, t) \in \mathcal{G} \Rightarrow (t, r_2, h) \in \mathcal{G}$.
- ☞ $\mathbf{h} + \mathbf{r}_1 = \mathbf{t}$ and $\mathbf{t} + \mathbf{r}_2 = \mathbf{h}$, this implies that $\mathbf{r}_2 \approx -\mathbf{r}_1$
- → Good for learning that "childOf" is just the "parentOf" vector pointed in the opposite direction.

Multi-relational Graph Embedding: TransE

- ☞ **Composition**: $(h, r_1, th), (th, r_2, t) \in \mathcal{G}$ and $\Rightarrow (h, r_3, t) \in \mathcal{G}$.
- $\rightarrow$ $\mathbf{h} + \mathbf{r}_1 = \mathbf{th}$ and $\mathbf{th} + \mathbf{r}_2 = \mathbf{t}$. This implies that $\mathbf{r}_3$ must be $\approx \mathbf{r}_1 + \mathbf{r}_2$
- ☞ **Transitive**: $(h, r, th), (th, r, t) \in \mathcal{G}$ and $\Rightarrow (h, r, t) \in \mathcal{G}$.
- $\rightarrow$ $\mathbf{h} + \mathbf{r} = \mathbf{th}$ and $\mathbf{th} + \mathbf{r} = \mathbf{t}$ This implies that $\mathbf{r}$ must be $\approx 2\mathbf{r}$ This only works if $\mathbf{r} = 0$.
- ☞ **1-to-N**: $(h, r, t_1) \in \mathcal{G} \Rightarrow \exists (h, r, t_2) \in \mathcal{G}$.
- $\rightarrow$ $\mathbf{h} + \mathbf{r} = \mathbf{t}_1, \mathbf{h} + \mathbf{r} = \mathbf{t}_2, \ldots$ This forces the embeddings for $\mathbf{t}_1, \mathbf{t}_2, \ldots$ to be identical points in the space. The representation of an entity is the same when involved in any relations, ignoring distributed representations.
- ☞ Despite its simplicity and efficiency, TransE has flaws in dealing with $1-\text{to}-N$ or $N-\text{to}-N$ relations (ex. *directorOf*)

Multi-relational Graph Embedding: TransE



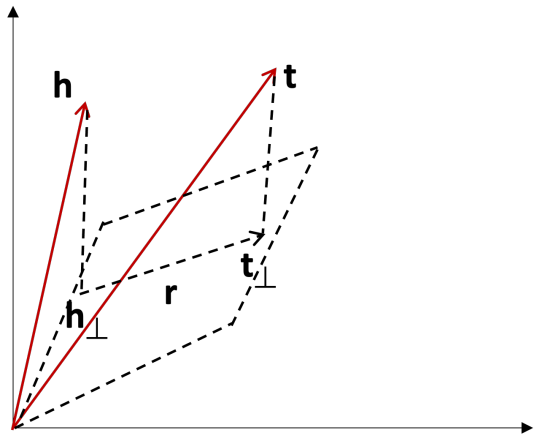
$\Rightarrow$

Multi-relational Graph Embedding: TranH

- The core idea behind TransH (Translating on Hyperplanes) is to allow an entity to have different representations depending on the relationship it is involved in. Specifically designed to fix TransE's inability to handle 1-to-N, N-to-N, and Reflexive relationships. TransH introduced relation-specific hyperplanes .
- An entity have different representations when involved in different relations. **h** and **t** are projected onto the hyperplane.
- Each **r** is a vector on a hyperplane with $\mathbf{w}_r$ the projection normal vector (which defines the orientation of the relation's hyperplane ($||w_r||_2 = 1$).

$$\mathbf{h}_\perp = \mathbf{h} - \mathbf{w}_r^T \mathbf{h} \mathbf{w}_r \text{ and } \mathbf{t}_\perp = \mathbf{t} - \mathbf{w}_r^T \mathbf{t} \mathbf{w}_r$$

- $f_r(h,t) = -||\mathbf{h}_\perp + \mathbf{r} - \mathbf{t}_\perp||_2^2$



Multi-relational Graph Embedding: TransR

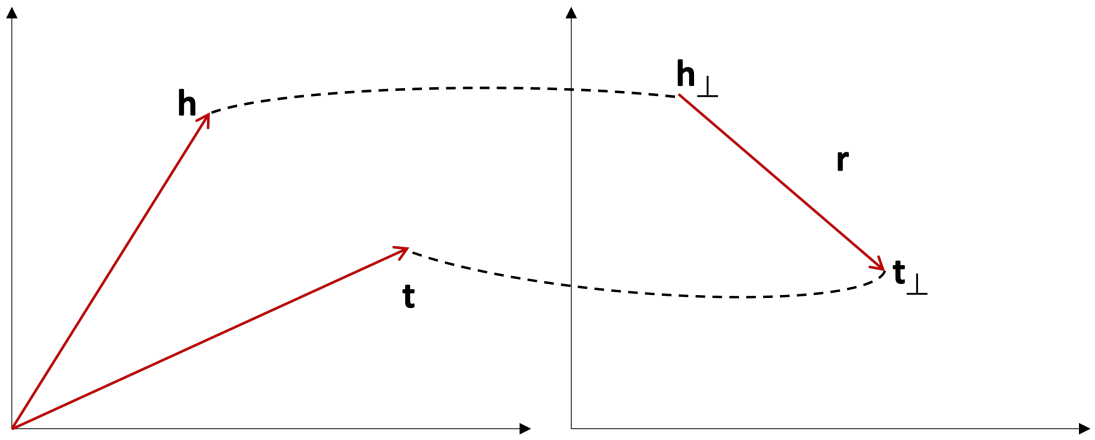
- TransR ((Translating in Relation Spaces) shares a very similar idea with TransH but introduces a relation specific spaces .
- In TransR, entities are represented as vectors in an entity space $\mathbb{R}^d$, and each relation is associated with a specific space $\mathbb{R}^k$, modeled as a translation vector in that space.
- The core idea is that an entity can have many facets, and each relationship only cares about a specific one.
- Powerful in modelling complex relations. it introduces a projection matrix for each relation $\mathcal{O}(dk)$ parameters for each relation whereas TransE and TransH require $\mathcal{O}(d)$ per relation.

Multi-relational Graph Embedding: TransR

- In TransR, entities are in $\mathbb{R}^d$, while every single relationship has its own unique Relation Space ($\mathbb{R}^k$
- Project: It takes $\mathbf{h}, \mathbf{t}$ from Entity Space and moves them into the specific space for relationship $\mathbf{r}$ using a projection matrix $\mathbf{M}_r$: $\mathbf{h}_\perp = \mathbf{M}_r\mathbf{h}$ and $\mathbf{t}_\perp = \mathbf{M}_r\mathbf{t}$ moves d to k
- Translate: $\mathbf{h}_\perp + \mathbf{r} \approx \mathbf{t}_\perp$

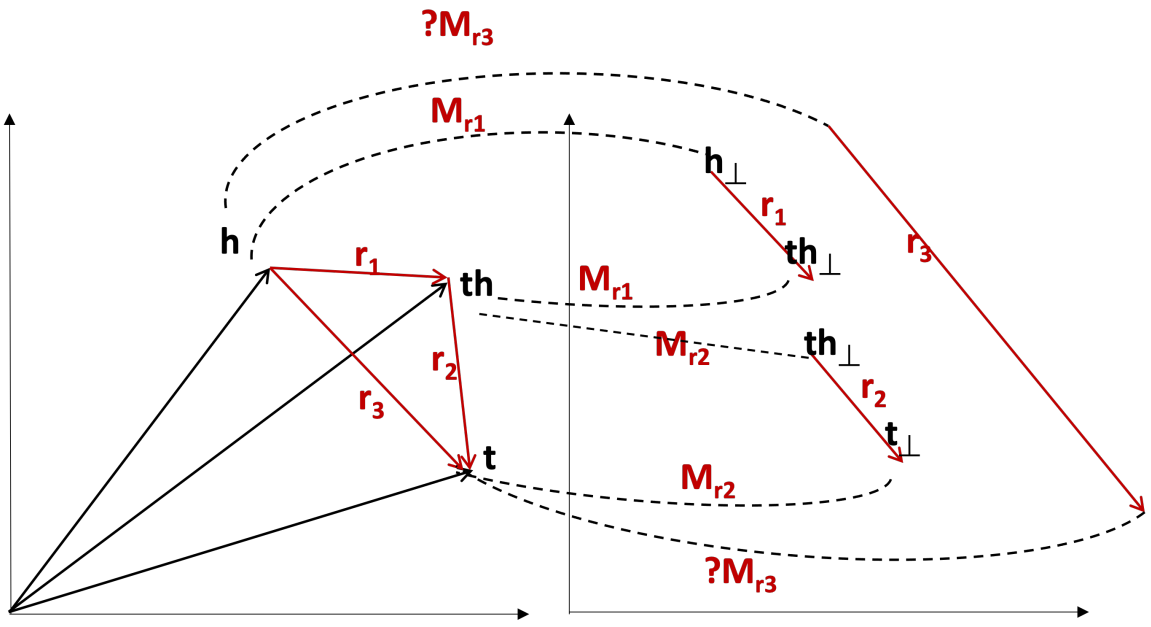
$$f_r(h,t) = ||\mathbf{h}_\perp + \mathbf{r} - \mathbf{t}_\perp||_2^2$$

where each $\mathbf{M}_r \in \mathbb{R}^{k \times d}$ is a projection matrix from the entity space to the relation space of r.



Multi-relational Graph Embedding: TransR

- $\mathbf{M}_r \neq \mathbf{M}_{-r}$ and $\mathbf{M}_{r_1} \neq \mathbf{M}_{r_2}$
- Composition: hard for TransR to satisfy $\mathbf{M}_{r_1}\mathbf{t} + \mathbf{r}_1 + \mathbf{r}_2 = \mathbf{M}_{r_2}\mathbf{h}$ because the projection matrices $\mathbf{M}_{r_1}$ and $\mathbf{M}_{r_2}$ are usually completely different.
- TransR* a specific variant where the matrices themselves are designed to be "composable," using $\mathbf{M}_{r_1} \times \mathbf{M}_{r_2}^{+}$ (Moore-Penrose Pseudoinverse) to represent the transition between the two spaces.



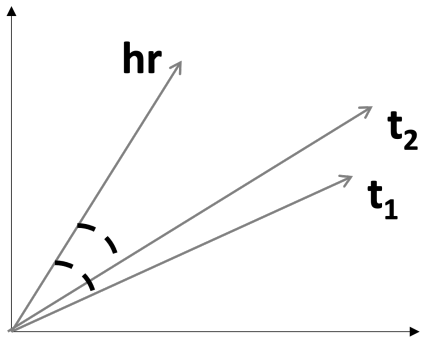
Multi-relational Graph Embedding: DistMult

- DistMult the simplest in this category uses bilinear-product-based models between $\mathbf{h}, \mathbf{r}$ and $\mathbf{t}$ to compute the scores.

$$f_r(h, t) = \sum_i \mathbf{h}_i \mathbf{r}_i \mathbf{t}_i$$

  It measures the similarity between the head and tail weighted by the relation (in the context of this relationship).

- True triple: The vectors align so that their product is a large positive number.
- False triple: The vectors are unaligned (or point in opposite directions), resulting in a low or negative score.
- DistMult "thinks" every relationship is a two-way street.



Multi-relational Graph Embedding: Summary

Model $f_r(h,t)$	**Sym.**	**Comp.**	**1-N**	**N-N**
TransE	$\times$	✓✓	$\times$	$\times$
TransH	✓	✓	✓	✓
TransR	✓✓	$\times$	✓✓	✓✓
DistMult	Always	✓	✓	✓

Method	Entity	Relation	scoring function	Regularisation	# parameters
TransE	$\mathbf{h},\mathbf{t} \in \mathbb{R}^d$	$\mathbf{r} \in \mathbb{R}^d$	$-\|\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|\|_2$	$\|\|\mathbf{h}\|\|_2 = 1, \|\|\mathbf{t}\|\|_2 = 1$	$\mathcal{O}(d(\|E\|+\|R\|))$
TransH	$\mathbf{h},\mathbf{t} \in \mathbb{R}^d$	$\mathbf{r},\mathbf{w}_r \in \mathbb{R}^d$	$-\|\|\mathbf{h}-\mathbf{w}_r^T\mathbf{h}\mathbf{w}_r+\mathbf{r}-\mathbf{t}-\mathbf{w}_r^T\mathbf{t}\mathbf{w}_r\|\|_2^2$	$\|\|\mathbf{h}\|\|_2 \leq 1, \|\|\mathbf{t}\|\|_2 \leq 1$ $\|\mathbf{w}_r^T\mathbf{h}\|/\|\|r\|\|_2 \leq \varepsilon$, $\|\|\mathbf{w}_r\|\|_2 \leq 1$	$\mathcal{O}(d(\|E\|+2\|R\|))$
TransR	$\mathbf{h},\mathbf{t} \in \mathbb{R}^d$	$\mathbf{r} \in \mathbb{R}^k$, $\mathbf{M}_r \in \mathbb{R}^{k \text{ x } d}$	$-\|\|\mathbf{M}_r\mathbf{h}+\mathbf{r}-\mathbf{M}_r\mathbf{t}\|\|_2^2$	$\|\|\mathbf{h}\|\|_2 \leq 1, \|\|\mathbf{t}\|\|_2 \leq, \|\|\mathbf{r}\|\|_2 \leq 1$ $\|\|\mathbf{M}_r\mathbf{h}\|\|_2 \leq 1$, $\|\|\mathbf{M}_r\mathbf{t}\|\|_2 \leq 1$	$\mathcal{O}(d(\|E\|+k\|R\|))$
DistMult	$\mathbf{h},\mathbf{t} \in \mathbb{R}^d$	$\mathbf{r} \in \mathbb{R}^d$,	$\sum_{i=1}^d \mathbf{r}_i\mathbf{r}_i\mathbf{t}_i$	$\|\|\mathbf{h}\|\|_2 \leq 1, \|\|\mathbf{t}\|\|_2 \leq, \|\|\mathbf{r}\|\|_2 \leq 1$	$\mathcal{O}(d)$

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Multi-relational Graph Embedding: Model perfomances

	Task	Example	Response
Link Prediction	Triple classification	(Joseph Fourrier, occupation, physicist) ?	(yes, 95%)
	Tail classification	(Joseph Fourrier, occupation, ?)	(1, physicist, 95%) (2, chemist, 93%)
	Head classification	(?, occupation, physicist)	(1, Albert Einstein, 91%), (2, Stephan Hawking, 89%)
	Relation prediction	(Joseph Fourrier, ?, physicist)	(1, occupation, 95%)
	Entity classification	(Joseph Fourrier, isTypeOf, ?)	(1, Person, 95%) (1, Scientist, 99%)

- ☞ Performance indicators are often used to measure the embedding quality of a model. Their simplicity makes them very suitable for evaluating the performance of an embedding algorithm even on a large scale. Given $\mathcal{Q}$ as the set of all ranked predictions of a model:
- ☞ Hits@K or H@K = $\frac{|\{q \in Q : q < k\}|}{|Q|} \in [0, 1]$ measures the probability to find the correct prediction in the first top K model predictions. Usually, used for $k = 10$, it reflects the accuracy of an embedding model to predict the relation between two given triples correctly.
  Larger values mean better predictive performances.
- ☞ Mean rank is the average ranking position of the triples predicted correctly by the model among all the possible triples. $MR = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} q$
  The smaller the value, the better the model.
- ☞ Mean reciprocal rank measures the number of triples predicted correctly. $MRR = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{q} \in [0, 1]$
  The larger the value, the better the model.

Pipeline for Link Prediction

- **Data Preparation & Preprocessing**: The triples are split into train, validation, and test sets. Ensure that all entities and relation types in the test set also appear in the training set (to avoid the "out-of-vocabulary" problem).
  Negative Sampling: "corrupting" triples randomly replacing the head or tail for each positive one.
- **Model Selection & Embedding Initialisation**: Usually initialised with random noise. Translational embeddings are a "structure-only" -based models.
  It treats entities and relation types as unique IDs (node_01, node_02), learning purely based on thei connections in KG.
- **Optimisation**: The model iteratively updates the embeddings to optimise the scoring function.
- **Test**: For every test triple (h, r, t): corrupt the triple by replacing the tail with every other possible entity e in the KG. Rank with or without corrupted.
  creating a complete "haystack" of candidates, then see where the "needle" (the true tail) ranks.
  If the true tail is at Rank 1, the model is perfect.
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Multi-relational and Knowledge Graphs

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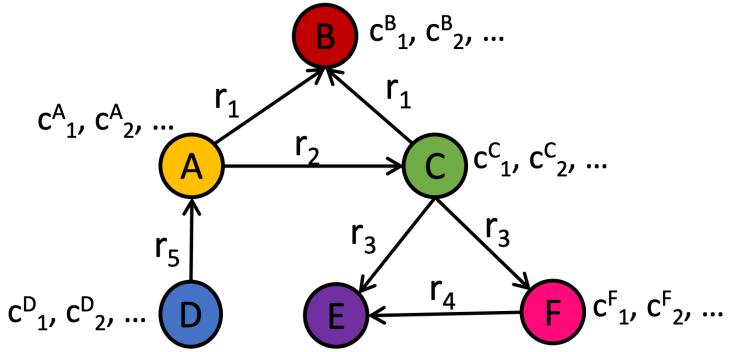
Nodes: entities of different classes c_1, c_2, ...
Edges: relationships between entities of different types: r_1, r_2, r_3, r_4 and r_5